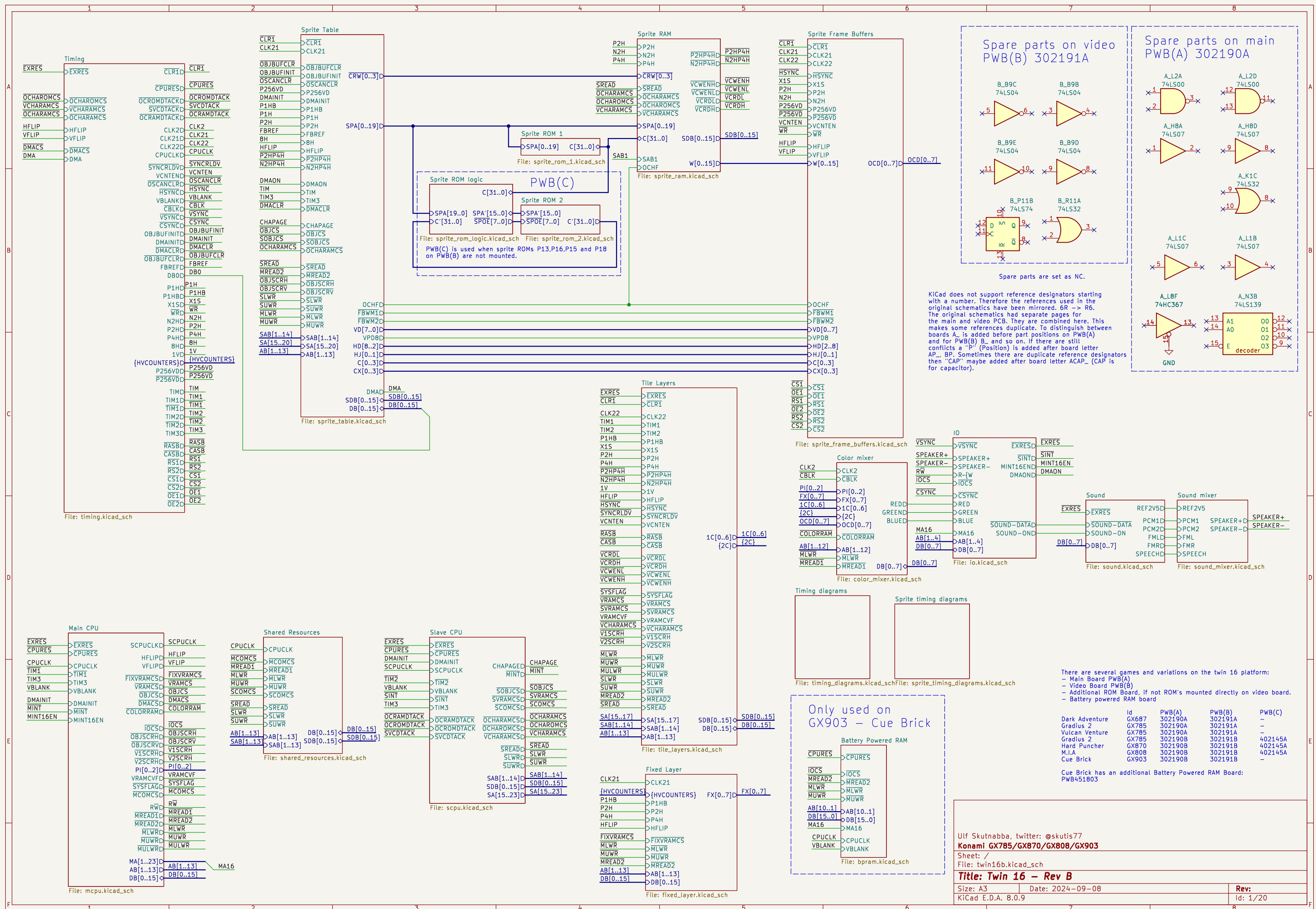
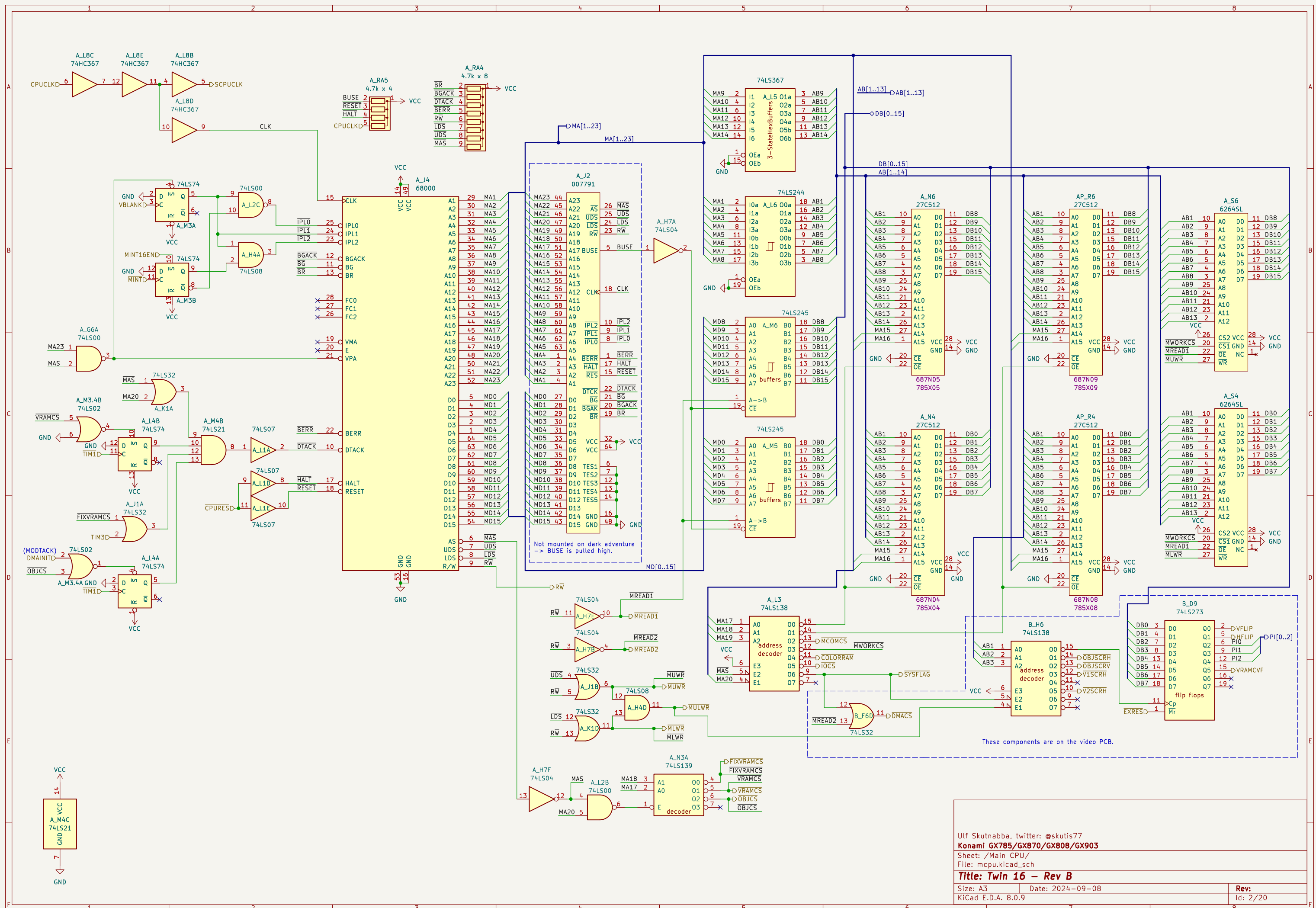
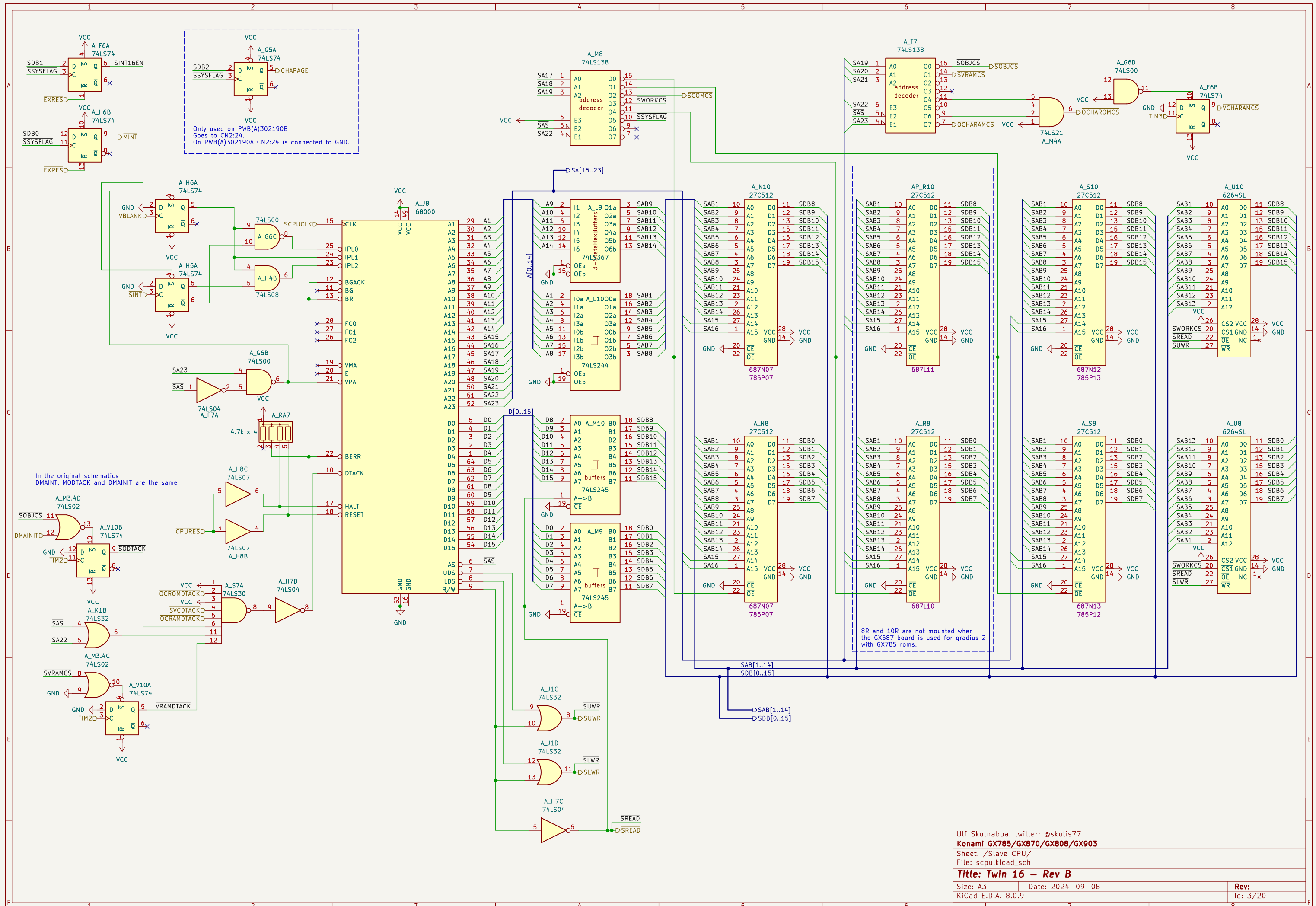




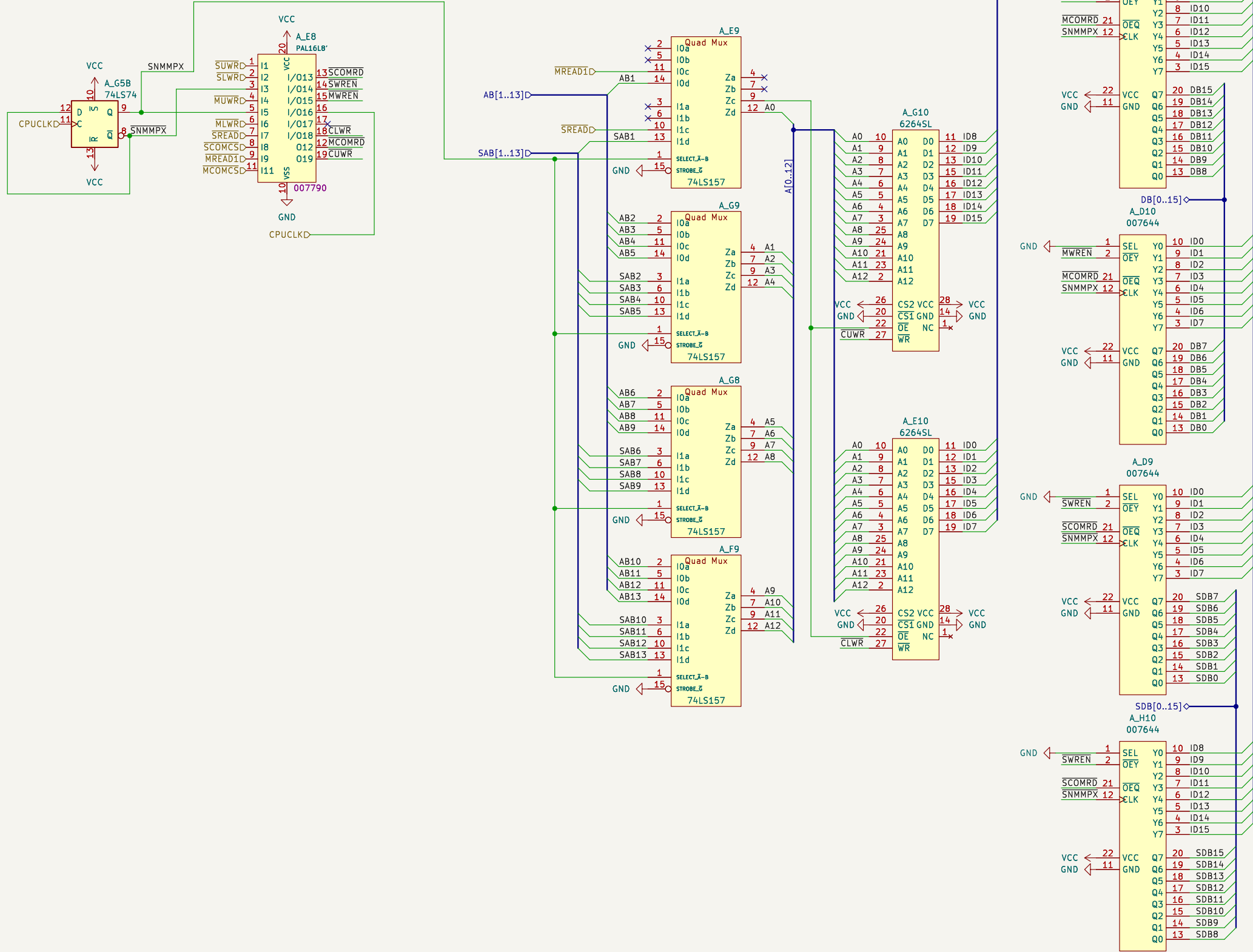




Konami 007790

```
MCOMRD = ~(-MREAD1 & ~MCOMCS) = MREAD1 | MCOMCS
SCOMRD = ~(-SREAD & ~SCOMCS)
SWREN = ~(-SNMMPX & SREAD & ~SCOMCS)
MWREN = ~(-SNMMPX & MREAD1 & ~MCOMCS)
CLWR = ~(-SLWR & ~SNMMPX & SNMMPX & SREAD & ~SCOMCS & ~CPUCLK | ~SNMMPX & ~MLWR & MREAD1 & ~MCOMCS & ~CPUCLK)
CUWR = ~(-SUWR & ~SNMMPX & SNMMPX & SREAD & ~SCOMCS & ~CPUCLK | ~MUWR & ~SNMMPX & MREAD1 & ~MCOMCS & ~CPUCLK)
```

All AND logic can be converted to OR form. But since AND logic and inverters are used in the PAL16L8 device it is kept in that form. The equations follow verilog syntax. The active low signals are not inverted, the bar is there only to show activeness.



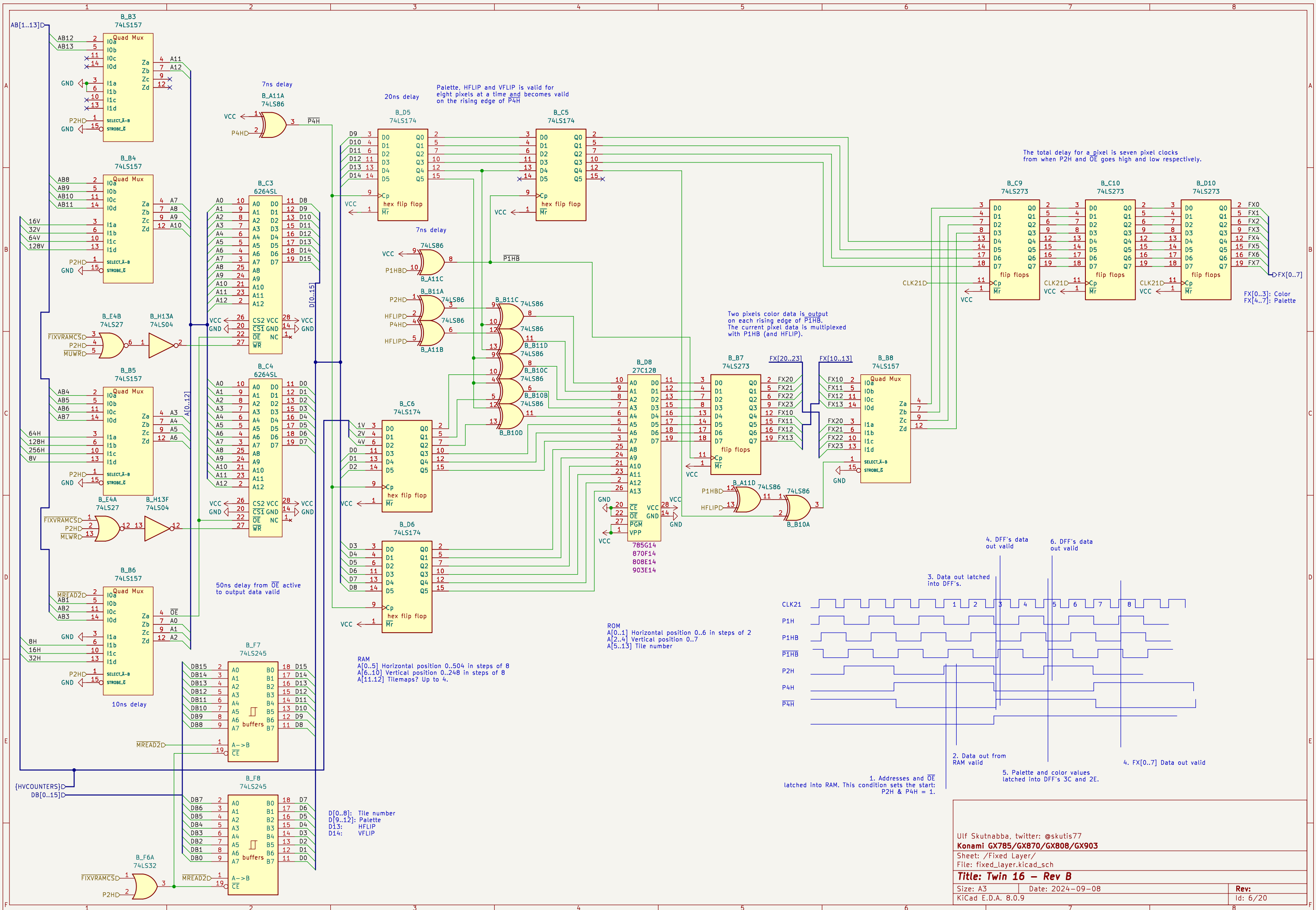
This is from jammarcade.net

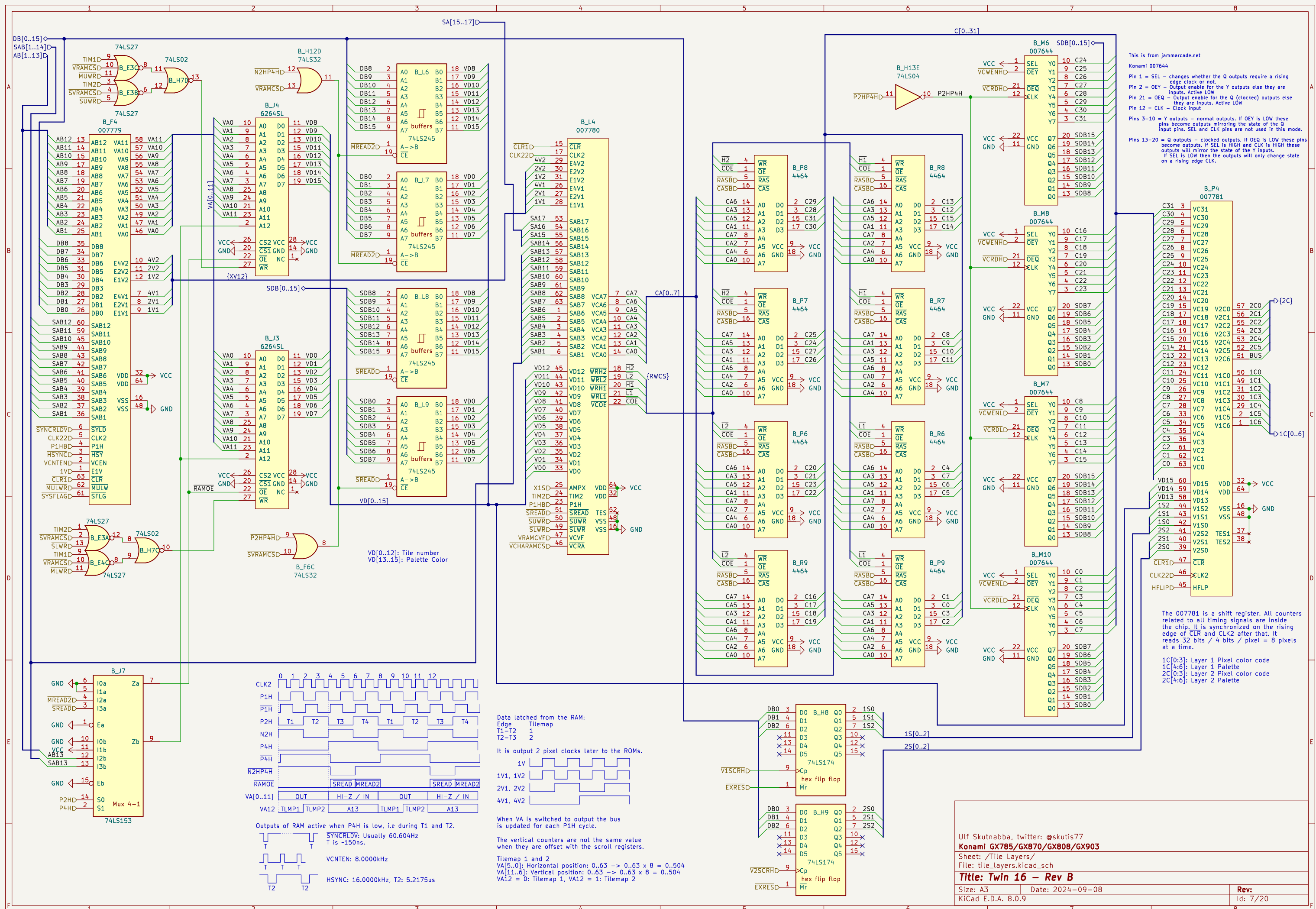
Pin 1 = SEL – changes whether the Q outputs require a rising edge clock or not.
Pin 2 = OEY – Output enable for the Y outputs else they are inputs. Active LOW
Pin 21 = OEQ – Output enable for the Q (clocked) outputs else they are inputs. Active LOW
Pin 12 = CLK – Clock input

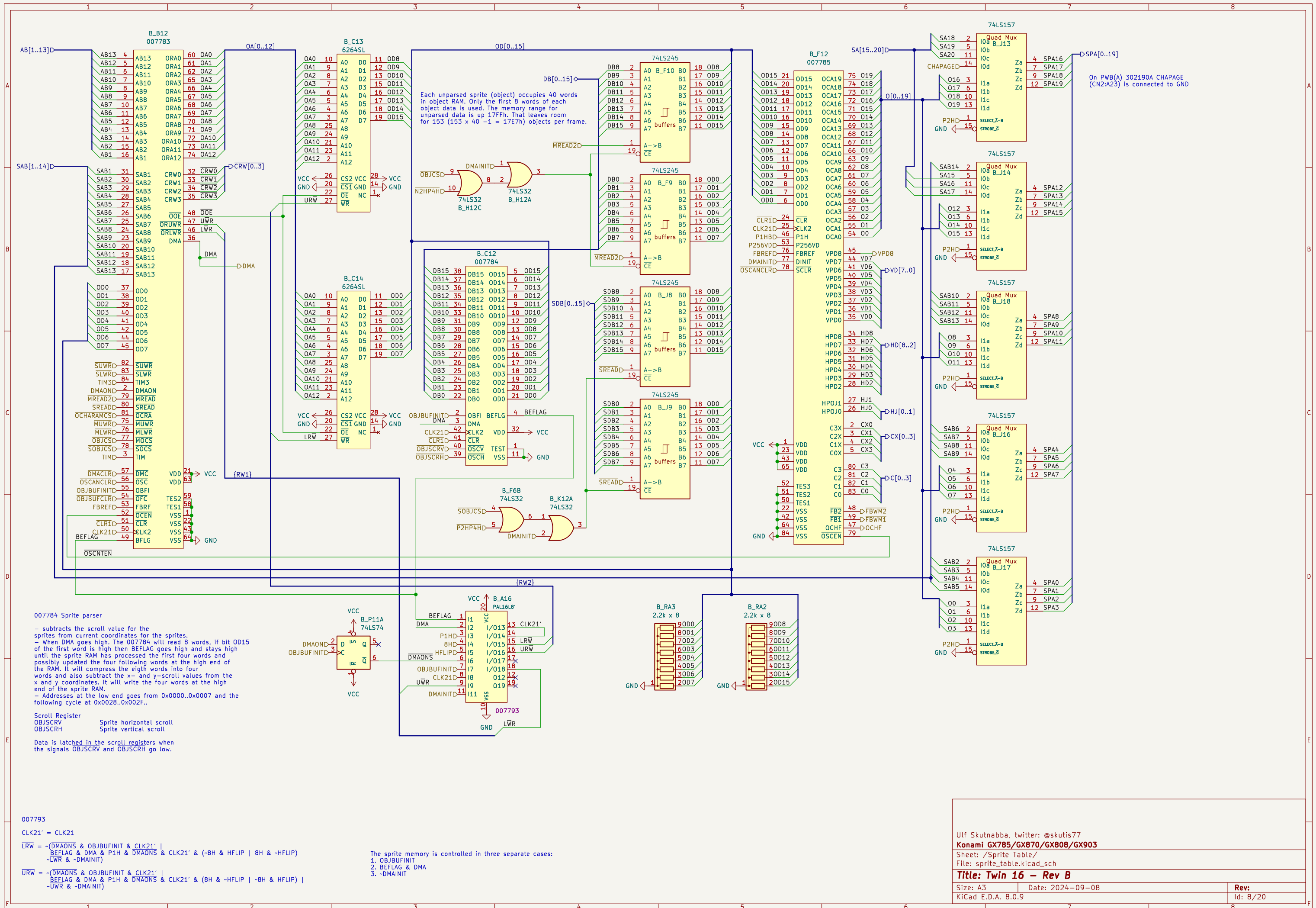
Pins 3–10 = Y outputs – normal outputs. If OEY is LOW these pins become outputs mirroring the state of the Q input pins. SEL and CLK pins are not used in this mode.

Pins 13–20 = Q outputs – clocked outputs. If OEQ is LOW these pins become outputs. If SEL is HIGH and CLK is HIGH these outputs will mirror the state of the Y inputs. If SEL is LOW then the outputs will only change state on a rising edge CLK.









Graphics ROM and RAM

ROM naming	ID	P13	P15	P16	P18
cuebrickj	GX903	NP	NP	NP	NP
miaj	GX808	808D15	NP	808D17	NP
hpuncher	GX870	870C15	870C16	870C17	870C18
gradius2 / vulcan	GX785	785F15	785F16	785F17	785F18

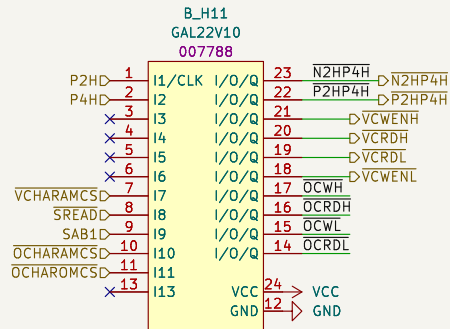
NP = Not Populated
On some boards P13,P15,P16 and P18 are NP and replaced with PWB(C) 402145A. See separate schematics.

This is from jammarcade.net

Pin 1 = SEL – changes whether the Q outputs require a rising edge clock or not.
Pin 2 = OEY – Output enable for the Y outputs else they are inputs. Active LOW
Pin 21 = OEQ – Output enable for the Q (clocked) outputs else they are inputs. Active LOW
Pin 12 = CLK – Clock input

Pins 3–10 = Y outputs – normal outputs. If OEY is LOW these pins become outputs mirroring the state of the Q input pins. SEL and CLK pins are not used in this mode.

Pins 13–20 = Q outputs – clocked outputs. If OEQ is LOW these pins become outputs. If SEL is HIGH and CLK is HIGH these outputs will mirror the state of the Y inputs. If SEL is LOW then the outputs will only change state on a rising edge CLK.



Konami 007788

The inverted -P2H is written as N2H.

OCRD1 = ~(-SREAD & SAB1 & ~OCHARAMCS & OCHAROMCS | ~SREAD & SAB1 & ~OCHAROMCS)
OCWL = ~(-P2H & SREAD & SAB1 & ~OCHARAMCS)
OCRDH = ~(-SREAD & ~SAB1 & ~OCHARAMCS & OCHAROMCS | ~SREAD & ~SAB1 & ~OCHAROMCS)
OCWH = ~(-P2H & SREAD & ~SAB1 & ~OCHARAMCS)
VCWENL = ~(-P2H & P4H & ~VCHARAMCS & SREAD & SAB1)
VCRDL = ~(-VCHARAMCS & ~SREAD & SAB1)
VCRDH = ~(-VCHARAMCS & ~SREAD & ~SAB1)
VCWENH = ~(-P2H & P4H & ~VCHARAMCS & SREAD & ~SAB1)
P2HP4H = ~(-P2H & P4H)
N2HP4H = ~(-P2H & P4H)

C[31..0]

Sprite Character Banks

Page	SPA19	SPA18	SPA17
1	X	L	L
2	X	L	L
3	H	H	L
4	X	H	L
5	RAM	X	H

SPA[0..19]D

CRW[0..3]D

B_H15C 74LS00

B_M15 SRAM32K8B

SPA0	10	A0	D0	11	C24
SPA1	9	A1	D1	12	C25
SPA2	8	A2	D2	13	C26
SPA3	7	A3	D3	15	C27
SPA4	6	A4	D4	16	C28
SPA5	5	A5	D5	17	C29
SPA6	4	A6	D6	18	C30
SPA7	3	A7	D7	19	C31
SPA8	25	A8			
SPA9	24	A9			
SPA10	21	A10			
SPA11	23	A11			
SPA12	2	A12			
SPA13	26	A13			
SPA14	1	A14			

B_M14 SRAM32K8B

SPA0	10	A0	D0	11	C16
SPA1	9	A1	D1	12	C17
SPA2	8	A2	D2	13	C18
SPA3	7	A3	D3	15	C19
SPA4	6	A4	D4	16	C20
SPA5	5	A5	D5	17	C21
SPA6	4	A6	D6	18	C22
SPA7	3	A7	D7	19	C23
SPA8	25	A8			
SPA9	24	A9			
SPA10	21	A10			
SPA11	23	A11			
SPA12	2	A12			
SPA13	26	A13			
SPA14	1	A14			

B_M19 SRAM32K8B

SPA0	10	A0	D0	11	C8
SPA1	9	A1	D1	12	C9
SPA2	8	A2	D2	13	C10
SPA3	7	A3	D3	15	C11
SPA4	6	A4	D4	16	C12
SPA5	5	A5	D5	17	C13
SPA6	4	A6	D6	18	C14
SPA7	3	A7	D7	19	C15
SPA8	25	A8			
SPA9	24	A9			
SPA10	21	A10			
SPA11	23	A11			
SPA12	2	A12			
SPA13	26	A13			
SPA14	1	A14			

B_M17 SRAM32K8B

SPA0	10	A0	D0	11	C0
SPA1	9	A1	D1	12	C1
SPA2	8	A2	D2	13	C2
SPA3	7	A3	D3	15	C3
SPA4	6	A4	D4	16	C4
SPA5	5	A5	D5	17	C5
SPA6	4	A6	D6	18	C6
SPA7	3	A7	D7	19	C7
SPA8	25	A8			
SPA9	24	A9			
SPA10	21	A10			
SPA11	23	A11			
SPA12	2	A12			
SPA13	26	A13			
SPA14	1	A14			

Pixel color data output rate is:
16-bits / 2 pixels = 8 bits / pixel.

W[15]: Vertical flip

B_P19 74LS374

C24	3	D0	00	2	W8
C25	4	D1	01	5	W9
C26	7	D2	02	6	W10
C27	8	D3	03	9	W11
C28	13	D4	04	12	W12
C29	14	D5	05	15	W13
C30	17	D6	06	16	W14
C31	18	D7	07	19	W15

3-state flip flops

N2H 11

OE 1

Cp 07

OE 07

OE 07

OE 07

OE 07

OE 07

OE 07

OE 07

OE 07

OE 07

OE 07

OE 07

OE 07

OE 07

OE 07

OE 07

OE 07

OE 07

OE 07

OE 07

OE 07

OE 07

OE 07

OE 07

OE 07

OE 07

OE 07

OE 07

OE 07

OE 07

OE 07

OE 07

OE 07

OE 07

OE 07

OE 07

OE 07

OE 07

OE 07

OE 07

OE 07

OE 07

OE 07

OE 07

OE 07

OE 07

OE 07

OE 07

OE 07

OE 07

OE 07

OE 07

Ulf Skutnabba, twitter: @skutis77

Konami GX785/GX870/GX808/GX903

Sheet: /Sprite RAM/

File: sprite_ram.kicad_sch

Title: Twin 16 – Rev B

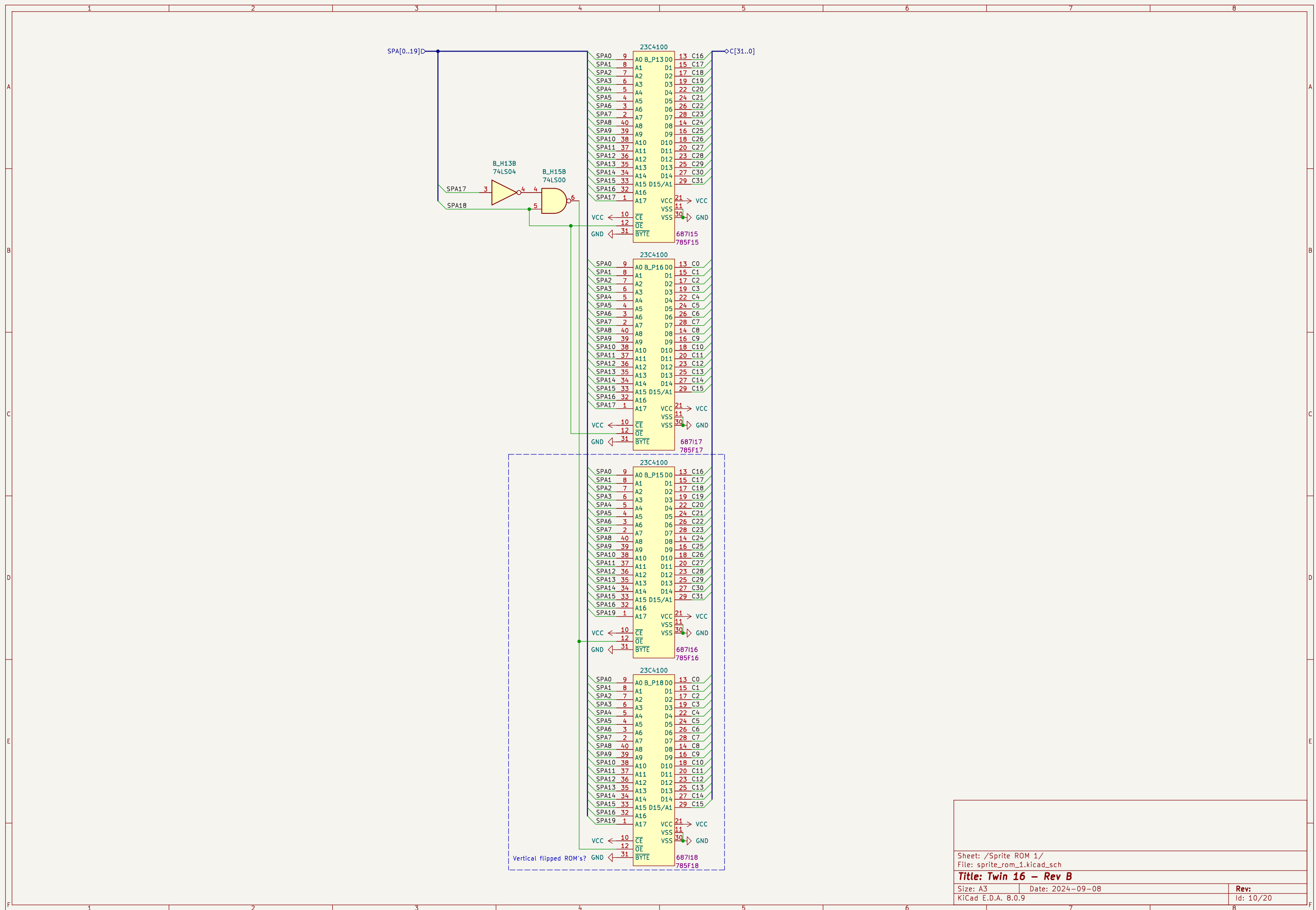
Size: A3

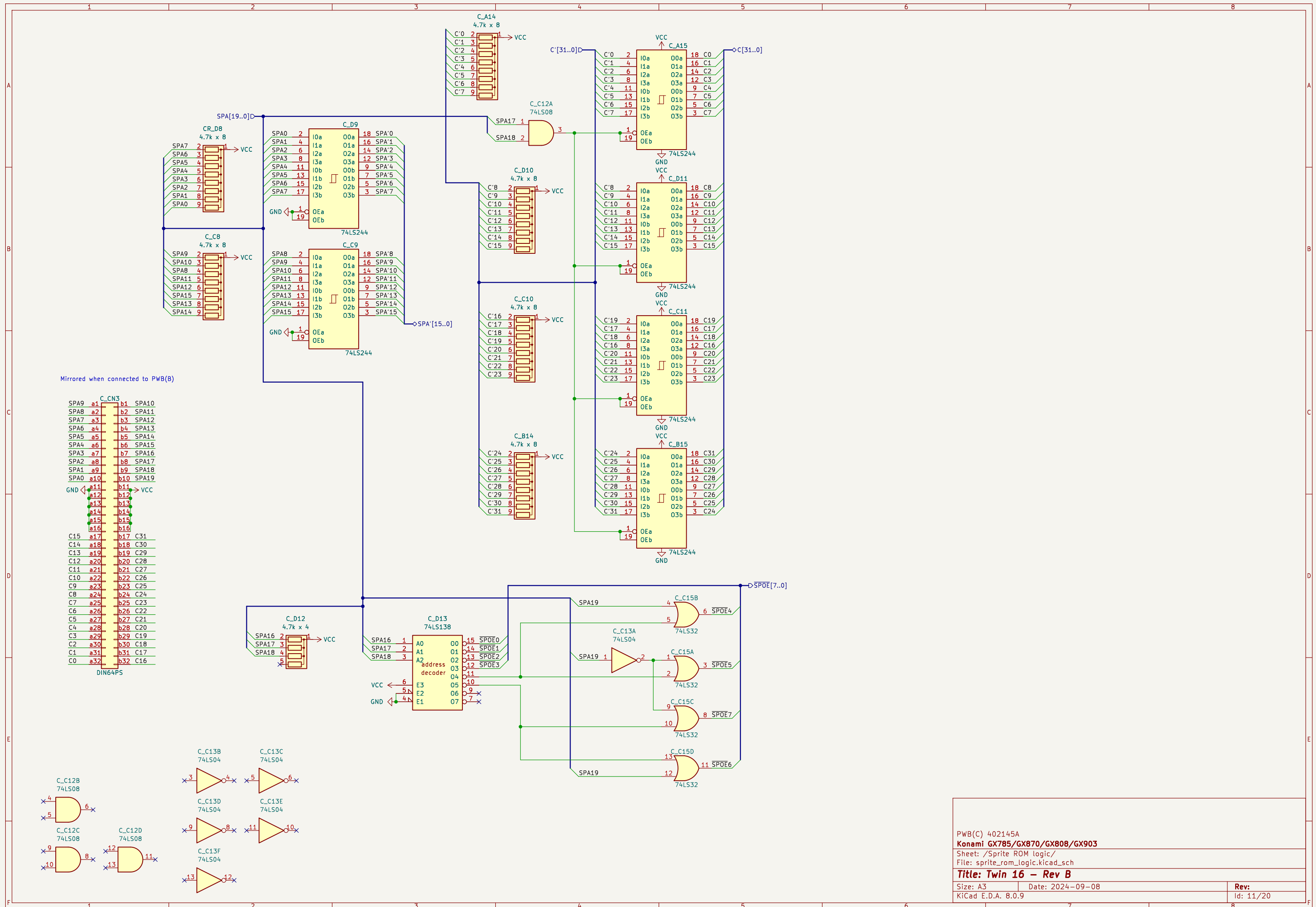
Date: 2024–09–08

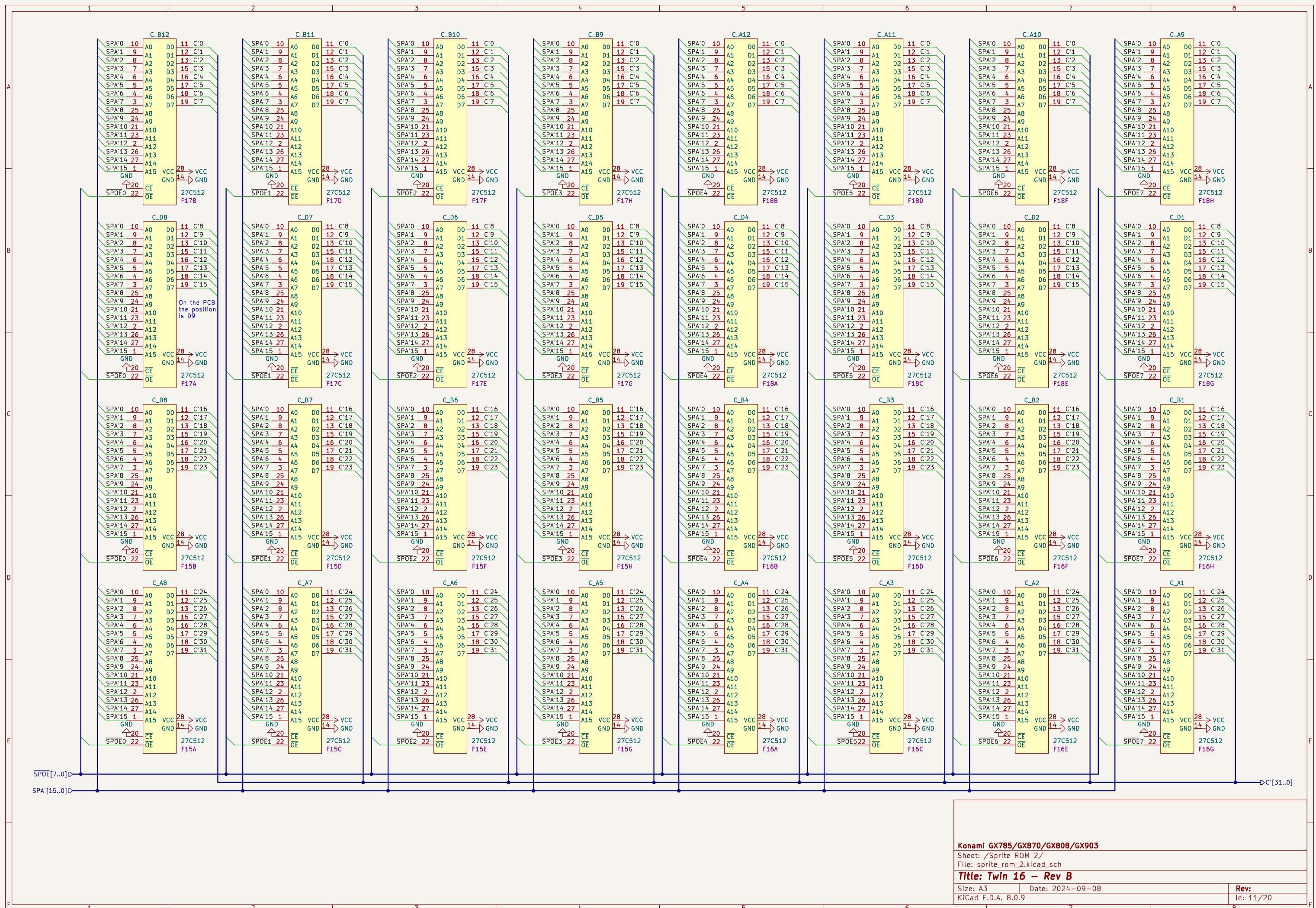
Rev:

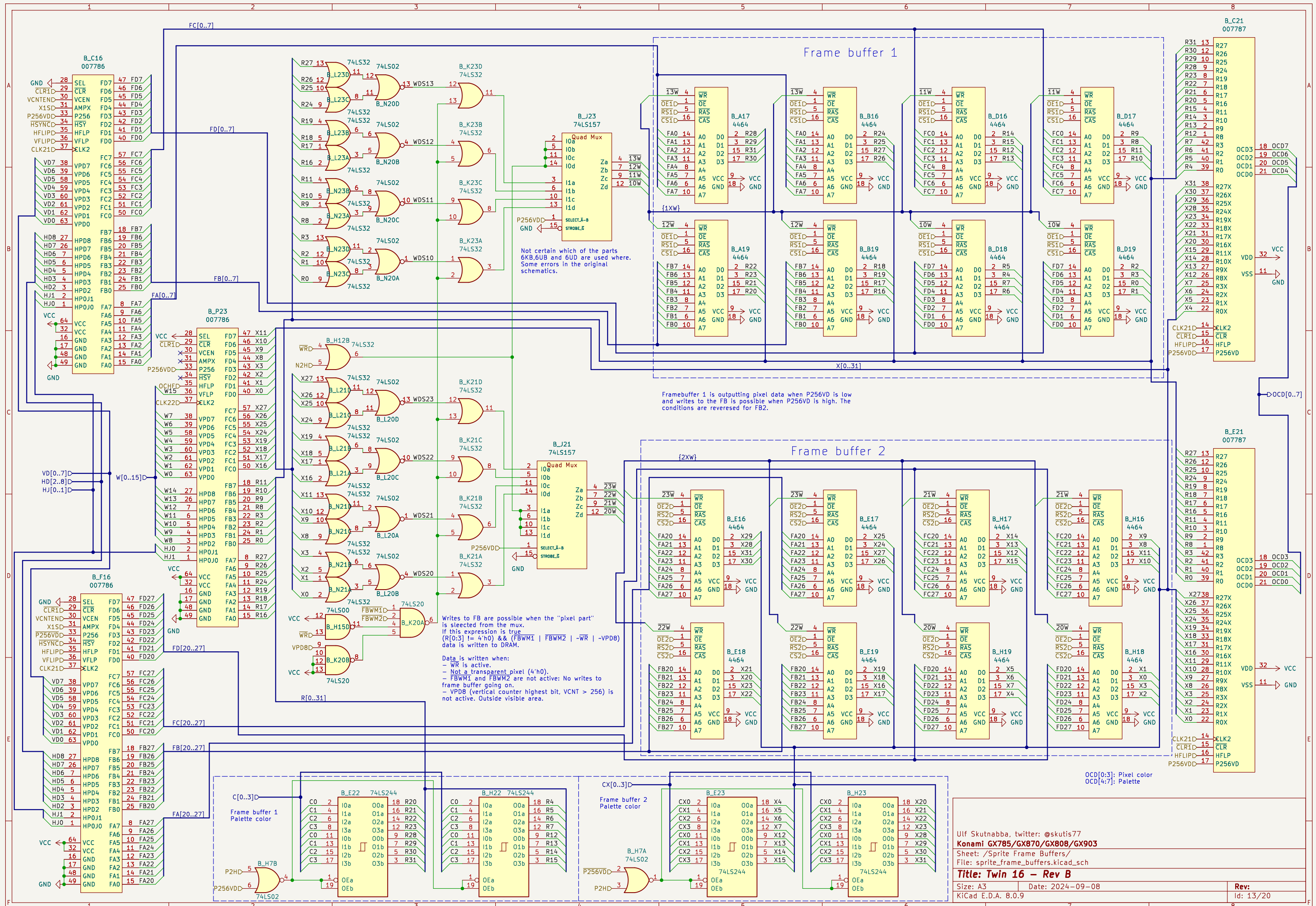
KiCad E.D.A. 8.0.9

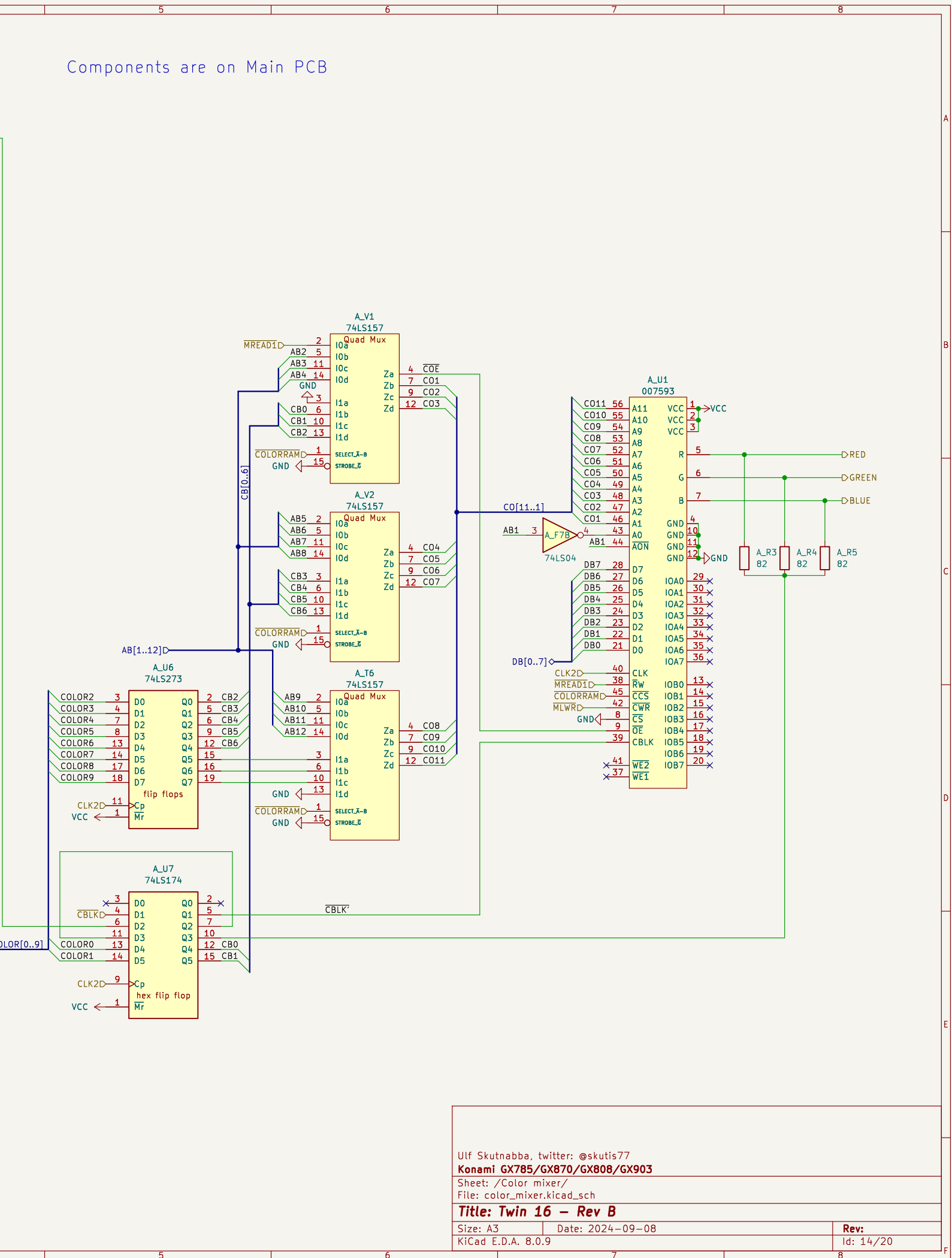
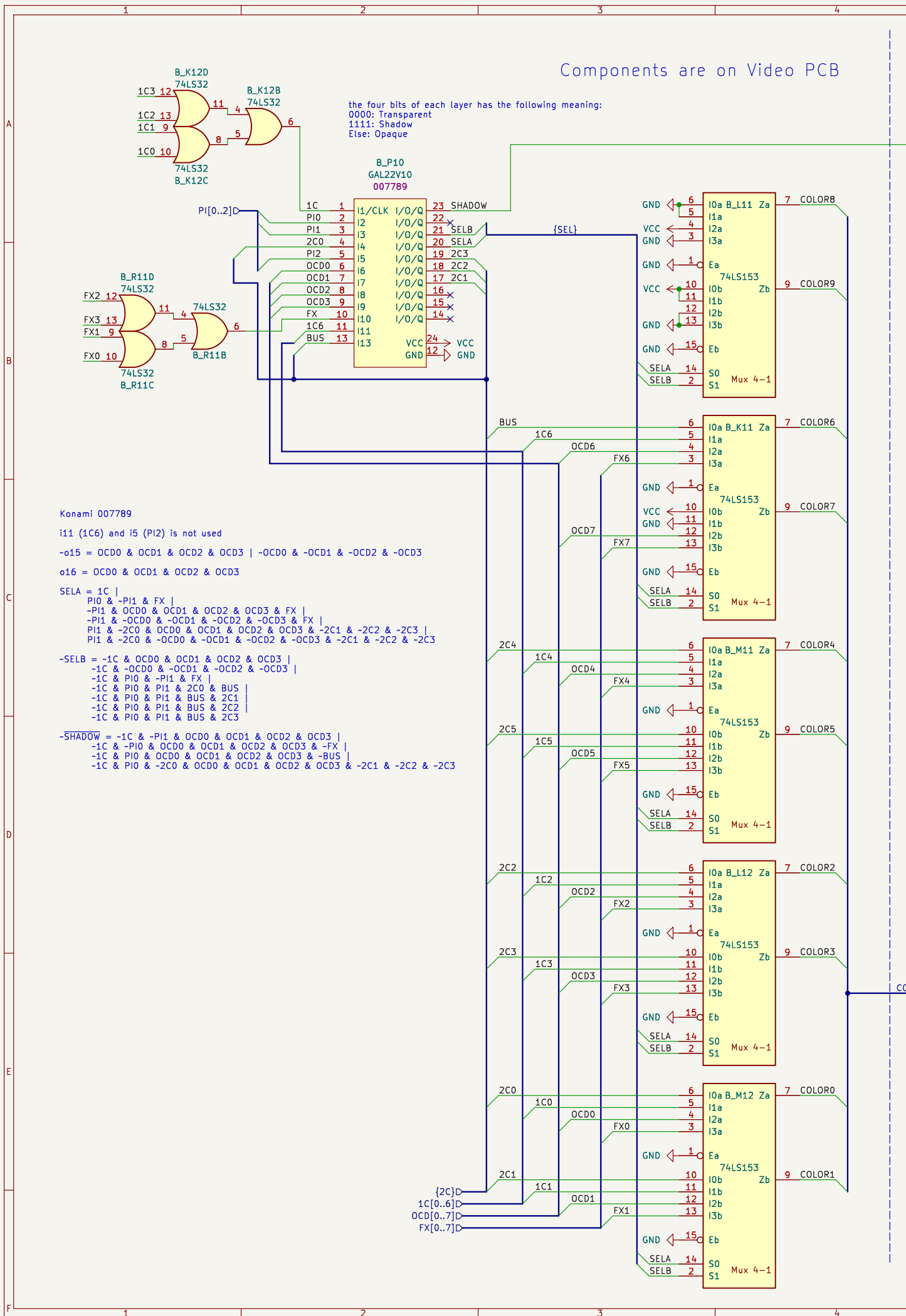
Id: 9/20







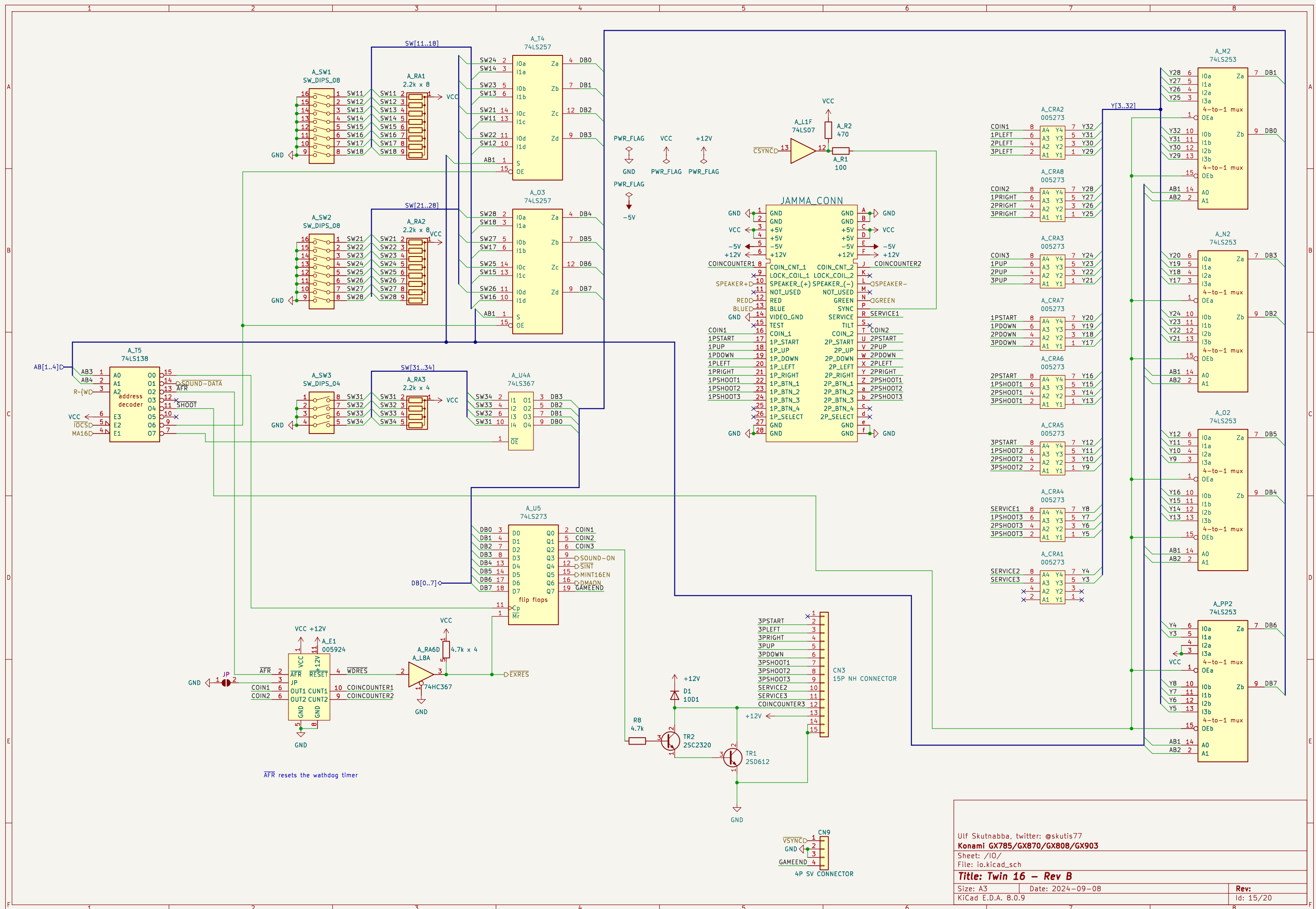


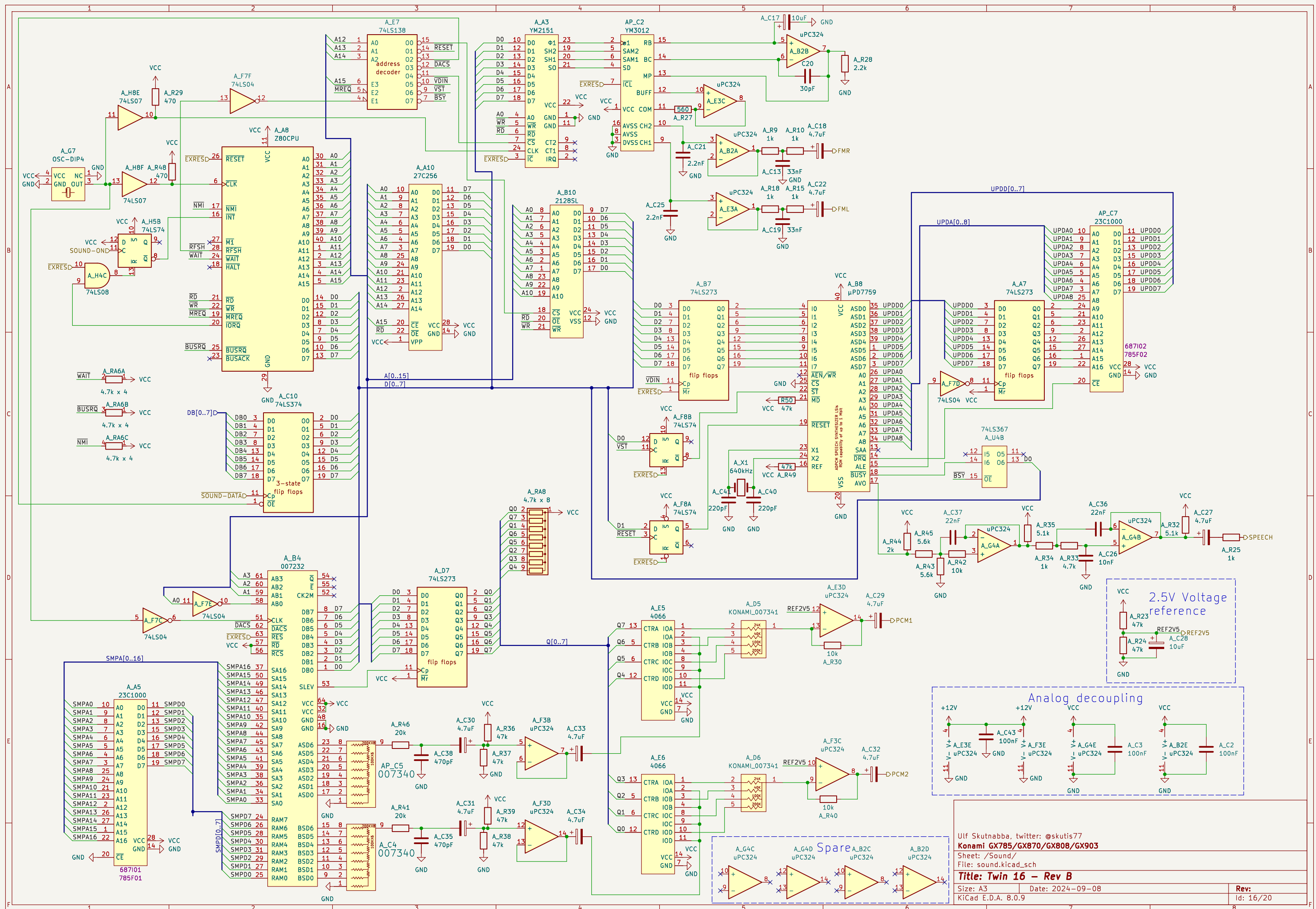


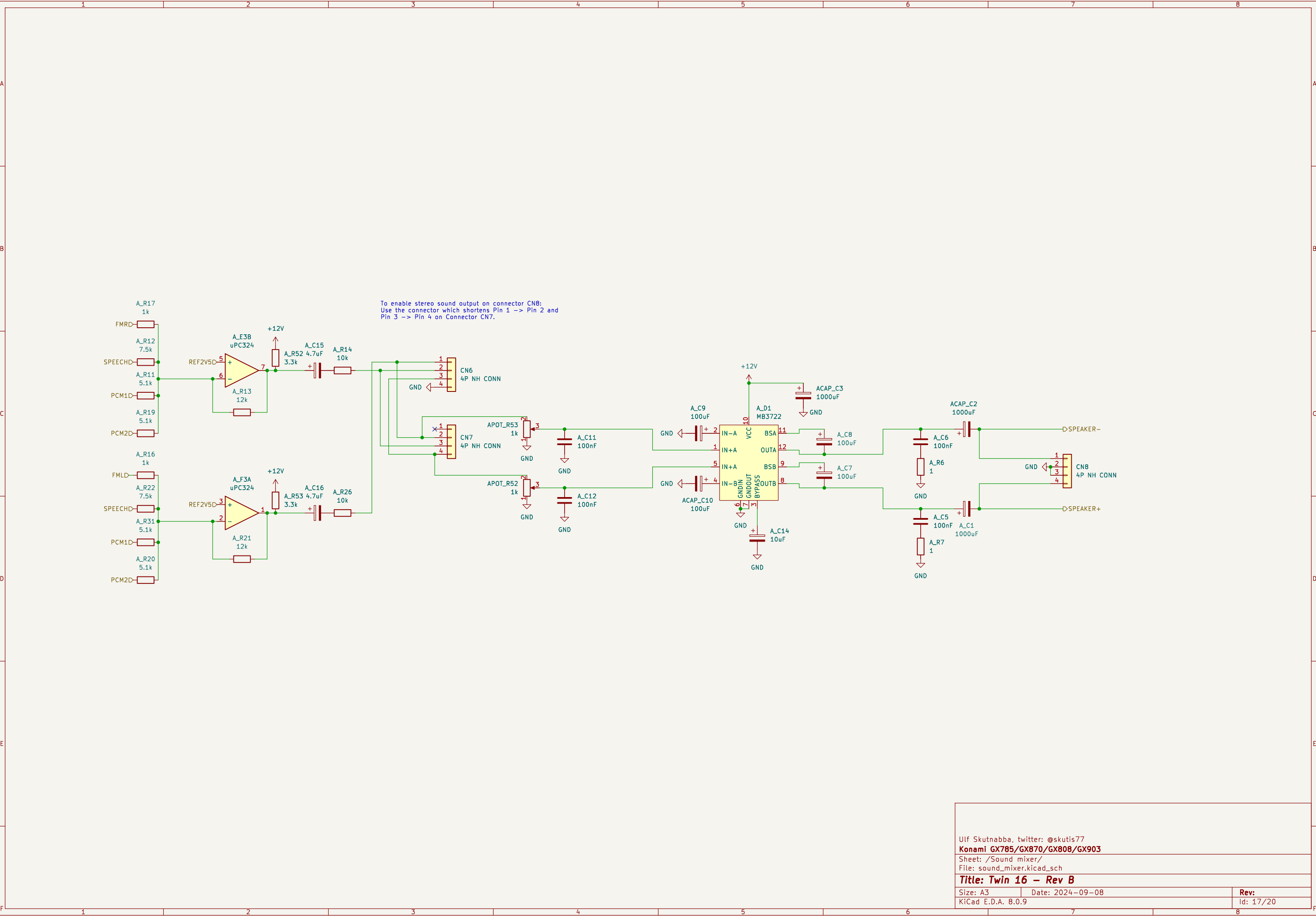
Size: A3	Date: 2024-09-08
KiCad E.D.A. 8.0.9	

7

Rev:
Id: 14/20







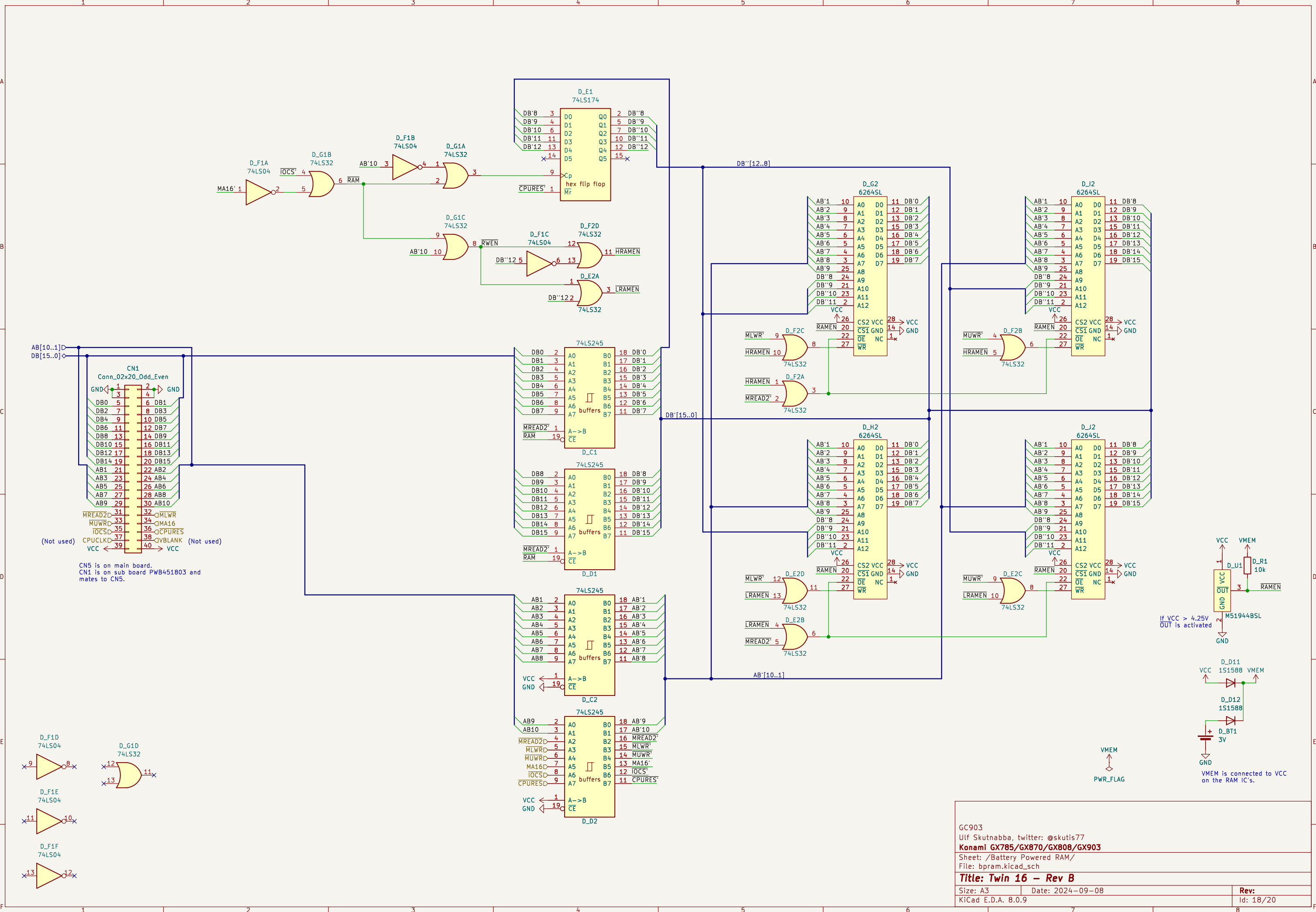
To enable stereo sound output on connector CN8:
Use the connector which shortens Pin 1 -> Pin 2 and
Pin 3 -> Pin 4 on Connector CN7.

Ulf Skutnabba, twitter: @skutis77
Konami GX785/GX870/GX808/GX903
Sheet: /Sound mixer/
File: sound_mixer.kicad_sch

Title: Twin 16 – Rev B

Size: A3 Date: 2024-09-08
KiCad E.D.A. 8.0.9

Rev:
Id: 17/20



Horizontal signals

$f = \text{CLK2} / 8 = 768\text{kHz}$
 $f = \text{CLK2} / 16 = 384\text{kHz}$
 $f = \text{CLK2} / 32 = 192\text{kHz}$
 $f = \text{CLK2} / 64 = 96\text{kHz}$
 $f = \text{CLK2} / 128 = 48\text{kHz}$
 $f = 2/3 \times \text{CLK2} / 128 = 32\text{kHz}$
 $f = 2/3 \times \text{CLK2} / 256 = 16\text{kHz}$
 $f = 2/3 \times \text{CLK2} / 256 = 16\text{kHz}$
 $f = 16\text{kHz} / 2 = 8\text{kHz}$
 $f = 16\text{kHz} / 2 = 8\text{kHz}$
 $f = 16\text{kHz} / 2 = 8\text{kHz}$

Signal	Description / Frequency
CLR	Reset signal
P4H	Pixel clock (768kHz)
8H	Horizontal clock (384kHz)
16H	Horizontal clock (192kHz)
32H	Horizontal clock (96kHz)
64H	Horizontal clock (48kHz)
128H	Horizontal clock (32kHz)
256H	Horizontal clock (16kHz)
HSYNC	Horizontal sync (16kHz)
HBLK (CBLK)	Horizontal blanking / Color clock (8kHz)
CSYNC	Color sync (8kHz)
VCNTEN	Vertical clock enable (8kHz)
1V	Vertical clock (8kHz)

The first HSYNC is at HCOUNT 175 after reset.

– The first VCNTEN is skipped after reset. It goes low 140ns before HSYNC goes low, and high again when HSYNC goes low. VCNTEN is active right before every second falling edge of HSYNC.

– CPURES goes high, and stays high, on the seventh falling edge of HSYNC.

The diagram illustrates the timing relationships for the U186:3 camera module. It shows the following signals and their characteristics:

- 1V**: $f = 16\text{kHz} / 2 = 8\text{kHz}$
- 2V**: $f = 16\text{kHz} / 4 = 4\text{kHz}$
- 4V**: $f = 16\text{kHz} / 8 = 2\text{kHz}$
- 8V**: $f = 16\text{kHz} / 16 = 1\text{kHz}$
- 16V**: $f = 16\text{kHz} / 32 = 500\text{Hz}$
- 32V**: $f = 16\text{kHz} / 64 = 250\text{Hz}$
- 64V**: $f = 16\text{kHz} / 128 = 125\text{Hz}$
- 128V**: $f = 16\text{kHz} / 264 = 60.606\text{Hz}$
- VSYNC**: Vertical sync signal.
- VBLANK**: Vertical blanking signal. High period is 7 x 16 x 2 = 224, Low period is 5 x 4 x 2 = 40.
- FBREF**: Frame rate reference signal. High period is 140ns, Low period is 12ns.
- SYNCRDLY**: Sync delay signal. Goes low 140ns before FBREF goes low, and high again when FBREF goes low.
- OBJBUFCLR**: Object buffer clear signal. Goes low one pixel clock before the rising edge of VBLANK and high again when VBLANK goes high.
- OSCANCRLR**: Oscillator clear signal. Goes low for 140ns - One pixel clock cycle.
- P256VD**: Power supply signal. Goes high/low one P4H cycle (8 pixels) after VSYNC goes high.

FBREF is measured at pin 22. At U186:3 FBREF is delayed by 22ns going high and 12ns going low.

The numbers in the HSYNC and HBLK diagrams are HSYNC cycles.
All edges are synchronised to the rising edge of CLK2.

Pixel clock CLK2 = 18.432MHz / 3 = 6.144MHz

Horizontal lines
HSYNC freq = 6.144MHz / 384 = 16kHz
HSYNC period = 1 / 16kHz = 62.5us
Pixels / line: 384
Active pixels: 320
Blank pixels: 64

Vertical lines
VSYNC freq = 16kHz / 264 = 60.606060Hz
VSYNC period = 1 / f = 264 / 16kHz = 16.5ms
scanlines / frame: 264
Active lines: 224
Blank lines: 40

The numbers in the VSYNC and VBLK diagrams are HSYNC cycles.
All edges are synchronised to the falling edge of HSYNC.

Front Porch (FP) 24 px.
Back Porch (BP) 8 px.

Front Porch (FP) and
Back Porch (BP) 16 HSYNC cycles.

CBLK'' is at the output of color mixer 007593.
CBLK' is shifted inside the 007593 one pixel clock.

HCOUNT is bits [256H, 128H, 64H, 32H, 16H, 8H, P4H, P2H, P1H]
VCOUNT is bits [128V, 64V, 32V, 16V, 8V, 4V, 2V, 1V]

Ulf Skutnabba, twitter: @skutis77
Konami GX785/GX870/GX808/GX903
Sheet: /Timing diagrams/
File: timing_diagrams.kicad_sch
Title: Twin 16 – Rev B
Size: A3 | Date: 2024-09-08 | Rev:
KiCad E.D.A. 8.0.9 | Id: 19/20

